Mid Examination-2 Tutorial Sheet (ECE/CSE)

Long Answer Questions

1. (a) State Cauchy's Mean Value Theorem and Verify Cauchy's mean value theorem for the functions $\sqrt{x}$ & $\frac{1}{\sqrt{x}}$ in (a, b)

(b) State Cauchy's Mean Value Theorem and Verify Cauchy's mean value theorem for the functions f(x)= logx and g(x)=x^2 in [a,b]

2.a) Expand log_e^x in powers of (x-1) and hence Evaluate $log_e^{1.1}$ Correct to 4 decimal places.

b) Obtain Maclaurin's series expansion of $Sinhx$

3.(a) If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$

(b) State Eulers theorem of Homogeneous function If $u = \log\left(\frac{x^2 + y^2}{x + y}\right)$ P. T $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ =1

4. Expand $x^2y + 3y - 2$ in powers of (x-1) and (y+2)

5. If $u = \frac{yz}{x}$, $v = \frac{xz}{y}$, $w = \frac{xy}{z}$ then prove that $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4$

6. If $u = x^2 - y^2$, $v = 2xy$ where $x = rcos\theta$, $y = rsin\theta$ $then\ S.T$ $\frac{\partial(u,v)}{\partial(r,\theta)} = 4r^3$

7. Verify whether $u = \frac{x+y}{1-xy}$, $v = \tan^{-1} x + \tan^{-1} y$ are functionally dependent or not. If so find the relation between them.

8. Discuss the max. and min. values of $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$

9.Find the maximum and minimum values of the function $x + y + z$ subject to $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$

10. Evaluate the double integral $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2)dxdy$

11. Evaluate $\iint_R y\,dx\,dy$, where R is a region in the first quadrant bounded by the parabolas y^2=4x and x^2=4y

12. Evaluate the double integral $\int_0^a \int_0^{\sqrt{a^2-y^2}} (x^2 + y^2)\ dx\ dy$ by changing to polar co-ordinates.

13. $Evaluate$ $\int\int r^3\ dr\ d\theta$ over the area bounded by the circles $r = 2sin\theta$ and $r = 4\sin\theta$

14. $Evaluate$ $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z}dxdydz.$

SHORT ANSWER QUESTIONS

1. Find ‘c’ of Cauchy’s mean value theorem for the functions x^2 & x^3 in $(1, 2)$

2. $Expand\ sinx$ in powers of $x - \frac{\pi}{4}$

3. Obtain Maclaurin’s series expansion of e^x.

4.Find the first and second partial derivatives of $z = x^3 + y^3 - 3axy$

5. State Euler’s theorem and if $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$, Prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$.

6. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if $z = x^2y - ysin(xy)$

7. State Maclaurin’s series of two variables

8.State Lagrange’s Undetermined Multipliers method

9. If $x = \frac{u^2}{v}, y = \frac{v^2}{u} find \frac{\partial(x,y)}{\partial(u,v)}$.

10. Evaluate $\int_0^2 \int_0^x y\,dy\,dx$

11.Evaluate $\int_0^{\pi} \int_0^{a} r^3 \sin\theta \cos\theta drd\theta$

12.Change the order of integration $\int_{-1}^{1} \int_0^{\sqrt{1-x^2}} f(x,y)dy\ dx$

13.. Evaluate $\int_0^1 \int_y^{\sqrt{y}} x^2y^2dxdy$

14. Evaluate $\int_0^1 \int_1^2 \int_2^3 xyzdxdydz$

15. Evaluate $\int_0^1 \int_0^2 \int_1^2 (x + y + z)dxdydz$